

Xiaoqian Liu

CONTACT INFORMATION	900 University Ave., Olmsted Hall 1349 Riverside, CA 92521	<i>E-mail:</i> xiaoqian.liu@ucr.edu <i>Website:</i> https://xiaoqian-liu.github.io/
RESEARCH INTERESTS	Methodology: Statistical Machine Learning, Computational Statistics, Numerical Optimization, Structured Estimation, Convex-Nonconvex Regularization, Robust Estimation, Transfer Learning Application: Evaluation of Large Language Models, Integrative Analysis of Multi-Omics Data, Transcriptome Deconvolution, Genetic Variation Annotation	
EMPLOYMENT	University of California, Riverside , Riverside, CA <i>Tenure-track Assistant Professor of Statistics</i> 2024/07 – present University of Texas MD Anderson Cancer Center , Houston, TX <i>Postdoctoral Fellow</i> 2022/08 – 2024/06 Argonne National Laboratory , Lemont, IL <i>Research Aide</i> 2021/08 – 2022/07 <i>Wallace Givens Associate</i> 2021/05 – 2021/08	
EDUCATION	North Carolina State University , Raleigh, NC Ph.D., Statistics 2018/08 – 2022/07 Renmin University of China , Beijing, China M.S., Statistics 2015/09 – 2018/06 China University of Mining and Technology , Xuzhou, China B.S., Mathematics and Applied Mathematics 2011/08 – 2015/06	
HONORS AND AWARDS	UCR Rho Sigma Rao Faculty Award, Rho Sigma Rao National Statistics Honorary Society 2026 UCR 2025-2026 Regents Faculty Fellowship, UCR Academic Senate 2025 IMS FSML 2025 Travel Award, The Institute of Mathematical Statistics (IMS) 2025 Best Poster Prize, 2023 SIAM Conference on Computational Science and Engineering 2023 UF Statistics 2023 Winter Workshop Travel Award, University of Florida 2023 2022 SDSS – Student & Early Career Travel Award, American Statistical Association 2022 Student Travel Award, North Carolina Chapter of the American Statistical Association 2020	
PUBLICATIONS	Note: the sign * at the beginning of a paper indicates alphabetical order of authorships; the sign [†] indicates co-first authorships; the sign ☒ denotes the corresponding author. <i>Peer-reviewed Publications</i> [1] Heng, Q. [†] ☒, Liu, X. [†] , and Chi, E. C. (2025). Anderson Accelerated Operator Splitting Methods for Convex-Nonconvex Regularized Problems. <i>IEEE Open Journal of Signal Processing</i> , 6, 1094-1108. [2] Liu, X. ☒, Han, X., Chi, E. C., and Nadler, B. (2025). A Majorization-Minimization Gauss-Newton Method for 1-Bit Matrix Completion. <i>Journal of Computational and Graphical Statistics</i> , 1-13. [3] Guo, S. [†] , Liu, X. [†] , Cheng, X. [†] , Jiang, Y., Ji, S., Liang, Q., ..., and Wang, W. ☒ (2025). A Deconvolution Framework that Uses Single-Cell Sequencing Plus a Small Benchmark Data Set for Accurate Analysis of Cell Type Ratios in Complex Tissue Samples. <i>Genome Research</i> , 35(1), 147-161. [Best Poster Award at The 2023 Leading Edge of Cancer Research Symposium].	

- [4] **Liu, X.** , Molstad, A. J., and Chi, E. C. (2023). A Convex-Nonconvex Strategy for Grouped Variable Selection. *Electronic Journal of Statistics*, 17(2), 2912-2961.
- [5] **Liu, X.** , Chi, E. C., and Lange, K. L. (2023) A Sharper Computational Tool for L_2 E Regression. *Technometrics*, 65(1), 117-126. **[Invited to present in the Technometrics session at The 65th Annual Fall Technical Conference].**
- [6] **Liu, X.** , and Chi, E. C. (2022). Revisiting Convexity-Preserving Signal Recovery with the Linearly Involved GMC Penalty. *Pattern Recognition Letters*, 156, 60-66.
- [7] **Liu, X.** , Vardhan, M., Wen, Q., Das, A., Randles, A., and Chi, E. C. (2021). An Interpretable Machine Learning Model to Classify Coronary Bifurcation Lesions. *The 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 4432-4435. IEEE.
- [8] Zhang, B. and **Liu, X.** (2019). Sparse Principal Component Analysis with Fused Penalty. *Statistical Research*, 36(4), 119–128.

Preprints / Manuscripts

- [9] Jiang, Y.[†], Montierth, M. D.[†], Yu, K.[†], Ji, S., Guo, S., Tran, Q., **Liu, X.**, ..., and Wang, W. (2025). Subclonal Mutation Load Predicts Survival and Response to Immunotherapy in Cancers with Low to Moderate Tumor Mutation Burden. Submitted. [bioRxiv 2024.07.03.601939](https://doi.org/10.1101/2024.07.03.601939).
- [10] **Liu, X.** , Yan, H., Shi, H., Montellier, E., Chi, E. C., Hainaut, P., and Wang, W. (2025). Transfer Learning for Survival-based Clustering of Predictors with an Application to *TP53* Mutation Annotation. Submitted. [bioRxiv 2025.10.06.680732](https://doi.org/10.1101/2025.10.06.680732)
- [11] Shi, H., Feng, Y., and **Liu, X.** (2026). Transfer Learning for Robust Structured Regression with Bi-level Source Detection. Submitted. [arXiv:2604.06659](https://arxiv.org/abs/2604.06659)
- [12] Qu, X., Heng, Q., Zeng, H., and **Liu, X.** (2026). An Interpretable and Scalable Framework for Evaluating Large Language Models. Submitted. [arXiv:2605.07046](https://arxiv.org/abs/2605.07046)

SOFTWARE

cBMM: Python implementation of an efficient computational framework for LLM evaluation.
TransL2E: R implementation of transfer learning for structured L2E regression.
TL-SCP: R implementation of the TL-SCP method for survival-based clustering of predictors.
MMGN: R / MATLAB implementations of the MMGN method for 1-bit matrix completion.
DeMixSC: R implementation of the DeMixSC framework for bulk RNA-seq deconvolution.
L2E: R package for robust structured regression via the L_2 criterion.
GMC: R package for variable selection via a convex-nonconvex regularization strategy.

PRESENTATIONS AND TALKS

Invited Talks

- [1] Transfer Learning for Robust Structured Regression with Bi-level Source Detection. *2026 Applied Statistics Symposium. June 16, 2026.*
- [2] Transfer Learning for Survival-based Clustering of Predictors with an Application to *TP53* Mutation Annotation. *2026 Biostatistics Symposium of Southern California. Jan. 22, 2026.*
- [3] Transfer Learning for Survival-based Clustering of Predictors with an Application to *TP53* Mutation Annotation. *ICSA China 2025 Conference. June 29, 2025.*
- [4] Transfer Learning for Survival-based Clustering of Predictors with an Application to *TP53* Mutation Annotation. *CMDB/GGB/MCBL Joint Seminar at University of California, Riverside. April 9, 2025.*
- [5] A Majorization-Minimization Gauss-Newton Method for 1-Bit Matrix Completion. *Econometrics Seminar, Department of Economics at University of California, Riverside. April 1, 2025.*

- [6] Transfer Learning for Survival-based Clustering of Predictors with an Application to *TP53* Mutation Annotation. *ICQMB Seminar at University of California, Riverside*. Feb. 25, 2025.
- [7] A Sharper Computational Tool for L_2E Regression. *The 65th Annual Fall Technical Conference*. Oct. 5, 2023.
- [8] A Convex-Nonconvex Strategy for Grouped Variable Selection. *The 36th New England Statistics Symposium (NESS)*. June 6, 2023.
- [9] A Convex-Nonconvex Strategy for Grouped Variable Selection. *Computational and Methodological Statistics (CMStatistics) 2022*. Dec. 19, 2022.

Tutorials and Workshops

- [10] A Tutorial on Boosting Methods. *Duke University (Randles Lab)*. Nov. 17, 2022.
- [11] R for Data Science. *Biomedical Data Science Workshop & Careers Panel, UCLA*. July 17, 2022.
- [12] A Tutorial on the CART Algorithm. *Duke University (Randles Lab)*. Nov. 9, 2021.

Contributed / Refereed Presentations

- [13] Transfer Learning for Robust Structured Regression with Bi-level Source Detection. *2026 Joint Statistical Meetings (JSM)*. Aug. 5, 2026.
- [14] Survival-based Clustering of Predictors in Cox Regression with an Application to *TP53* Mutation Annotation. *2024 Joint Statistical Meetings (JSM)*. Aug. 8, 2024.
- [15] Survival-based Clustering of Predictors in Cox Regression with an Application to *TP53* Mutation Annotation. *NCI Spring School on Algorithmic Cancer Biology (SSACB) 2024*. April 3, 2024.
- [16] A Convex-Nonconvex Strategy for Grouped Variable Selection. *Eastern North American Region (ENAR) 2023 Spring Meeting*. Mar. 21, 2023.
- [17] A Convex-Nonconvex Strategy for Grouped Variable Selection. *2022 Symposium on Data Science & Statistics (SDSS)*. June 9, 2022.
- [18] Randomized Projections in Derivative-Free Optimization. *Summer Argonne Student Symposium (SASSy) 2021*. July 30, 2021.

Poster Presentations

- [19] Transfer Learning for Survival-based Clustering of Predictors with an Application to *TP53* Mutation Annotation. *IMS 2025 FSML Workshop*. August 2, 2025.
- [20] Annotating *TP53* Mutations via Survival-based Clustering of Predictors in Cox Regression. *REACH24 – The 7th International LFS Association Symposium*. Oct. 20, 2024.
- [21] Survival-based Clustering of Predictors in Cox Regression with an Application to *TP53* Mutation Annotation. *American Association for Cancer Research (AACR) Annual Meeting 2024*. April 8, 2024.
- [22] Annotating *TP53* Germline Mutations from Patient Time-to-Cancer Diagnosis via Homogeneity Pursuit in Cox Regression. *The 2023 Leading Edge of Cancer Research Symposium*. Nov. 16, 2023.
- [23] A Majorization-Minimization Gauss-Newton Method for 1-Bit Matrix Completion. *Statistical Foundations of Data Science and their Applications: A Conference in Celebration of Jianqing Fan's 60th Birthday*. May 9, 2023.
- [24] A Majorization-Minimization Gauss-Newton Method for 1-Bit Matrix Completion. *UF Statistics 2023 Winter Workshop*. Jan. 13, 2023.

- [25] An Interpretable Machine Learning Model to Classify Coronary Bifurcation Lesions. *The 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. Oct. 31 – Nov. 4, 2021.
- [26] Revisiting Convexity-Preserving Signal Recovery with the Linearly Involved GMC Penalty. *International Chinese Statistical Association (ICSA) 2020 Applied Statistics Symposium*. Dec. 14, 2020.
- [27] Revisiting Convexity-Preserving Signal Recovery with the Linearly Involved GMC Penalty. *The 2020 Women in Statistics and Data Science (WSDS) Virtual Conference*. Oct. 1, 2020.

TEACHING EXPERIENCE

University of California, Riverside, Riverside, CA

Instructor

- STAT 160A (Elements of Probability and Statistical Theory) Fall 2024, 2025
- STAT 160B (Elements of Probability and Statistical Theory) Winter 2025, 2026
- STAT 209 (Software Tools for Big Data Analysis) Spring 2025, 2026

Rice University, Houston, TX

Guest Lecturer

- STAT423/623 (Probability in Bioinformatics and Genetics) Spring 2024
Topic: Regularized Likelihood Models in Bioinformatics

North Carolina State University, Raleigh, NC

Teaching Assistant

- ST779 (Advanced Probability for Statistical Inference) Spring 2022
- ST517 (Applied Statistical Methods) Fall 2021
- ST370 (Probability and Statistics for Engineers) Fall 2018, Spring 2019, Fall 2019

MENTORING AND ADVISING

PhD Advisor

- Xinhao Qu, PhD student in Statistics at UC Riverside 2026/02 - present
- Jordyn Niemiec, PhD student in GGB at UC Riverside 2026/03 - present

UC LEADS Faculty Mentor

- Kora Dey, Undergraduate in Statistics at UC Riverside 2026/01 - present

University Honors Capstone Project Mentor

- Swaraj Dash, Undergraduate in Data Science at UC Riverside 2025/01 - 2026/06
- Tista Palmukhopadhyay, Undergraduate in Statistics at UC Riverside 2025/01 - 2026/06

Research Mentor

- Rishabh Rai, Undergraduate in Actuarial Science at UC Riverside 2024/12 - 2025/05

Collaborative Project Advisor

- Haoming Shi, PhD student at Rice University 2023/01 - present
- Hao Yan, PhD student at UTHealth 2023/09 - 2025/10
- Lisa Lin, Undergraduate at Rice University (now PhD at Yale University) 2021/10 - 2022/12

PhD Qualifying Exam Committee

- Sitong Pan, PhD student in Statistics at UC Riverside 2026
- Zhuo Huang, PhD student in Statistics at UC Riverside 2026
- Yusen Zhang, PhD student in Computer Science at UC Riverside 2026
- Xun Wang, PhD student in Statistics at UC Riverside 2026
- Jingjing Zhang, PhD student in Statistics at UC Riverside 2026
- Emily Ouyang, PhD student in Statistics at UC Riverside 2026
- Liming Zhao, PhD student in Statistics at UC Riverside 2026
- John Pleines, PhD student in Statistics at UC Riverside 2025

PROFESSIONAL SERVICES	<i>Journal Reviewer</i>	
	- Journal of Computational and Graphical Statistics - Nature Communications - Technometrics - Statistics and Computing - Statistical Learning and Data Science - Journal of Statistical Computation and Simulation - Journal of Data Science - Communications in Statistics - Simulation and Computation - Genetics - PLOS Genetics	
	<i>Conference Reviewer</i>	
	- STAI-X 2026	
	<i>Department Service</i>	
	- Department Colloquium Committee	2025 - 2026
	- Department Award Committee	2025 - 2026
	- Graduate Program Committee	2024 - 2025
	- Graduate Admission Committee	2024 - 2025
	<i>Other Services</i>	
	- Reviewer for 2026 SLDS Student Paper Competition. - Session organizer at CFE-CMStatistics 2025. - Reviewer for 2025 SLDS Student Paper Competition. - Session chair at ENAR 2023 Spring Meeting. - Judge for ENAR 2023 Spring Meeting Poster Competition. - Session chair at 2022 Symposium on Data Science & Statistics.	
VOLUNTEER AND LEADERSHIP	Member, Stats Up AI Alliance	2024/01 – 2025/12
	- Work as a technical team member with cofounders to encourage and empower statisticians to fit in and embrace AI research. • Build and maintain the website of Stats Up AI, including organizing data resources, collecting review articles from domain fields, and maintaining social media accounts.	
	Volunteer, Alternative Intercultural Service Break , NCSU	2019/03 – 2019/03
	- Worked as a volunteer with ABCCM in Black Mountain, NC, including homeless services, gardening, and environmental protection services. - Visited and gave presentations at Black Mountain middle and elementary schools to introduce international cultures.	
	President, University Youth Volunteers Association, CUMT	2013/06 – 2014/06
	- Organized collaborative volunteer activities among local commonweal organizations in Xuzhou. - Organized the inaugural University Volunteer Forum with five universities and colleges in Xuzhou.	